

StatTesting Instructions

Last Updated: 6/14/2025 by Clara Shin (clara.shin@wsu.edu)

This is the **fourth step** of chemotaxis analysis in I2Tumbles. StatTesting.m performs statistical testing across multiple groups. Each group corresponds to one CellTrackAnalysis_results.xlsx file.

Step 0: Prepare Excel Files

1. Important: Change the name of CellTrackAnalysis_results.xlsx every time after running CellTrackAnalysis.m. For example, add the experimental group names after each run like below:

 CellTrackAnalysis_results_1DMSO.xlsx	Today at 6:05 PM	10 KB	Micros...k (.xlsx)
 CellTrackAnalysis_results_CB.xlsx	Today at 5:54 PM	13 KB	Micros...k (.xlsx)
 CellTrackAnalysis_results_L-Ser.xlsx	Today at 6:04 PM	19 KB	Micros...k (.xlsx)

Step 1: StatTesting Parameters

1. Open StatTesting.m
2. Adjust parameters.

Parameter Name	Value Range	Data Type	Description
test_type	1, 2, 3, 4	Integer	When test_type = 1: Performs two-sample KS test (kstest2) When test_type = 2: Performs Kruskal-Wallis test (kruskalwallis) When test_type = 3: Performs two-sample t-test (ttest2) When test_type = 4: Performs one-way anova (anova1)
group_count	> 0	Integer	Number of experimental groups
excel_column_name		String	Column to read from the PerTrackMetrics sheet. Examples are TumbleCount, Tumbles_per_s, MeanRunDuration_s, etc.

Step 2: Interpret Results

- Example of KS Test

```
>> StatTesting
Test selected : Two-sample KS Test (kstest2)
Please select 2 Excel file(s), one per group.
  Group 1 → CellTrackAnalysis_results_CB.xlsx
  Group 2 → CellTrackAnalysis_results_L-Ser.xlsx
  Group 1: 131 valid values read.
  Group 2: 260 valid values read.

Data matrix built: 260 rows x 2 groups.

--- Two-sample KS Test (kstest2) ---
Column tested : Tumbles_per_s
Groups       : Group1, Group2

KS statistic  : 0.402466
p-value      : 0.000000
Reject H0    : true
Significance  : *** (p < 0.001)

--- Summary Statistics: Tumbles_per_s ---
Group      N      Mean      Median      Std      IQR
-----
Group1     131     0.6297     0.5714     0.5864     0.9091
Group2     260     0.2022     0.0000     0.3256     0.3922
```

- Example of ANOVA

```
>> StatTesting
Test selected : One-way ANOVA (anova1)
Please select 3 Excel file(s), one per group.
  Group 1 → CellTrackAnalysis_results_CB.xlsx
  Group 2 → CellTrackAnalysis_results_L-Ser.xlsx
  Group 3 → CellTrackAnalysis_results_1DMSO.xlsx
  Group 1: 131 valid values read.
  Group 2: 260 valid values read.
  Group 3: 84 valid values read.

Data matrix built: 260 rows x 3 groups.

--- One-way ANOVA (anova1) ---
Column tested : Tumbles_per_s
Groups       : Group1, Group2, Group3

p-value      : 0.000000

--- Post-hoc Multiple Comparisons (Tukey-Kramer) ---
Significance  : *** (p < 0.001)

--- Summary Statistics: Tumbles_per_s ---
Group      N      Mean      Median      Std      IQR
-----
Group1     131     0.6297     0.5714     0.5864     0.9091
Group2     260     0.2022     0.0000     0.3256     0.3922
Group3      84     0.8525     0.8422     0.6267     0.8523
```

References

- [MATLAB: kstest](#)
- [MATLAB: kruskalwallis](#)
- [MATLAB: ttest2](#)
- [MATLAB: anova1](#)